

An Automated Pipeline for Flagging Outdated Citations in AI-Recommended Literature

Abstract

Large language models (LLMs) are increasingly used to search, summarize, and recommend scientific literature. Retrieval-augmented generation (RAG) has improved the ability of AI systems to ground generated answers in external sources, but citation quality remains a challenge. A citation may be genuine and relevant while still being outdated because newer studies have extended, contradicted, corrected, or superseded the cited work. This paper proposes an automated pipeline for identifying such citations. The proposed framework combines citation extraction, bibliographic verification, publication-date analysis, semantic retrieval of newer studies, citation-context analysis, citation-network information, and supersession detection. Rather than treating publication age as sufficient evidence of obsolescence, the system evaluates citations according to their research context and purpose. A risk score is proposed to classify citations as current, foundational but old, potentially outdated, or likely superseded. Existing work on citation recommendation, scientific claim verification, RAG, and large scholarly corpora provides the technical foundation for the approach. The paper also discusses evaluation strategies, limitations, research gaps, and future directions for trustworthy AI-assisted scholarly communication.

Keywords: citation verification, outdated citations, artificial intelligence, large language models, RAG, scientific literature, citation recommendation, bibliometrics, scholarly NLP.

1. Introduction

The rapid growth of scientific publications makes literature discovery increasingly difficult for researchers. Artificial intelligence, particularly large language models (LLMs), can reduce this burden by retrieving publications, summarizing research, and generating citation-supported answers. Retrieval-augmented generation (RAG) was introduced to combine language-model knowledge with information retrieved from an external collection, improving access to factual information and providing a mechanism for updating knowledge [1]. ([arXiv](#))

However, retrieving a legitimate paper does not guarantee that it is an appropriate

citation for a current research claim. Scientific knowledge changes continuously. New experiments may contradict previous findings, new datasets may replace older benchmarks, and improved methodologies may supersede earlier approaches. Therefore, a citation can be authentic and relevant but nevertheless **outdated for the specific claim being made**.

This problem differs from citation hallucination. Citation hallucination occurs when an AI system generates a nonexistent or incorrectly attributed source. In contrast, an outdated citation refers to a real publication but fails to represent the most appropriate current evidence.

For example, a 2018 paper may correctly introduce a machine-learning method. If an AI system states that the method is currently state-of-the-art while citing only the 2018 paper, the reference is historically correct but insufficient for the current claim.

This paper proposes an automated pipeline that evaluates citation currency using multiple signals rather than publication age alone. The main objectives are:

1. verifying the identity and status of cited publications;
2. determining citation age and disciplinary context;
3. retrieving newer related literature;
4. identifying evidence of improvement or supersession;
5. analyzing citation intent; and
6. producing an interpretable outdated-citation warning.

2. Background and Related Work

2.1 Citation Recommendation

Citation recommendation has traditionally focused on retrieving publications that are relevant to a document or citation context. Bhagavatula et al. developed a content-based citation recommendation approach that combines document representations with candidate reranking [2]. Färber and Jatowt surveyed citation-recommendation methods and highlighted the importance of document context, retrieval methods, datasets, and evaluation strategies [3].

These approaches are valuable for identifying relevant literature, but relevance does not necessarily imply currency. A recommendation model can successfully retrieve a historically important paper while overlooking more recent evidence.

2.2 Citation Intent

The purpose of a citation is another important consideration. Cohan et al. developed SciCite, which classifies citations according to their rhetorical intent [4]. For example, a citation may provide background information, describe a methodology, or support a comparison. Citation intent is important because older papers are not necessarily inappropriate. A researcher citing the original paper that introduced a method may legitimately use an old reference. In contrast, a claim about the **current state of the art** requires recent evidence.

2.3 Scientific Claim Verification

Scientific claim verification provides an important foundation for the proposed system. Wadden et al. introduced SciFact, a dataset and task for determining whether scientific abstracts support or refute scientific claims [5]. The work demonstrates that automated systems can retrieve evidence and reason about scientific statements. ([ACL Anthology](#))

SciFact-Open extended this problem to a substantially larger open-domain scientific corpus and showed that systems trained on smaller collections can experience significant performance degradation when applied to large-scale literature [6]. ([arXiv](#))

The proposed system extends this idea by adding a temporal dimension:

[
Claim $\rightarrow$ Evidence $\rightarrow$ Newer Evidence
]

The question becomes not only whether a paper supports a claim, but whether newer evidence changes the status of that support.

2.4 Scholarly Corpora

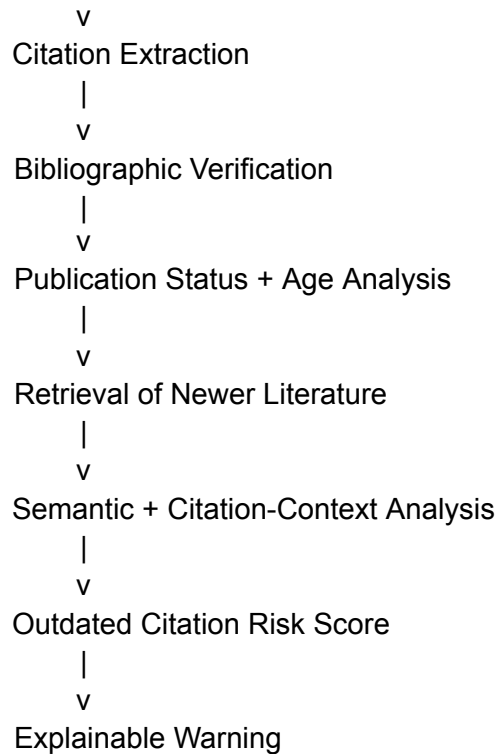
Large scholarly corpora are necessary for automated citation analysis. S2ORC provides millions of academic papers with metadata, abstracts, citation links, and structured full text [7]. ([arXiv](#)) Such resources allow systems to analyze citation contexts and construct large-scale scholarly graphs.

3. Proposed Automated Pipeline

The proposed system consists of seven major stages:

AI-Generated Answer

|



3.1 Citation Extraction

The system first identifies all citations contained in an AI-generated response. For each citation, it stores:

- citation marker;
- bibliographic information;
- surrounding sentence;
- paragraph;
- associated claim.

Preserving the citation context is essential because the same paper can be appropriate for one claim and outdated for another.

3.2 Bibliographic Verification

Each citation is matched against a scholarly database using identifiers such as DOI, title, author, year, and venue.

A matching score can be represented as:

```
[  
M=w_1D+w_2T+w_3A+w_4Y  
]
```

where (D), (T), (A), and (Y) represent DOI, title, author, and year agreement.

Citations that cannot be reliably resolved should be reported as **unresolved citations**, rather than automatically classified as outdated.

3.3 Temporal Analysis

The simplest temporal feature is publication age:

$$\begin{aligned} &[\\ &\text{Age} = Y_{\{\text{current}\}} - Y_{\{\text{publication}\}} \\ &] \end{aligned}$$

However, age alone is insufficient. A 30-year-old theoretical paper may remain foundational, while a three-year-old empirical result may become outdated quickly.

Therefore, the system should consider:

- discipline;
- citation purpose;
- research topic;
- publication rate;
- availability of newer evidence.

3.4 Retrieval of Newer Literature

For each older citation, the system searches for papers

published afterward. Candidate papers are retrieved using:

- keyword similarity;
- title and abstract similarity;
- scientific embeddings;
- shared datasets;
- shared methods;
- citation relationships.

Semantic similarity can be represented using cosine similarity:

$$\begin{aligned} &[\\ &\text{Sim}(A,B) = \\ &\frac{A \cdot B}{\|A\| \|B\|} \\ &] \end{aligned}$$

A high similarity score indicates that two papers address related topics, but it does **not** automatically demonstrate that the newer paper

supersedes the older one.

3.5 Supersession Detection

The system searches newer papers for evidence that an earlier approach has been improved, replaced, or contradicted.

Relevant signals include phrases such as:

- "improves upon";
- "outperforms";
- "extends";
- "limitations of";
- "previous methods";
- "new state of the art."

A supersession score can be defined as:

$$[S_{sup} = \alpha Sim + \beta E + \gamma P]$$

where:

- (Sim) = semantic similarity;
- (E) = explicit comparison evidence;
- (P) = performance-improvement evidence.

3.6 Citation-Intent Analysis

The system classifies the purpose of each citation.

Citation type Need for recent evidence

Historical Low

Foundational theory Low

Original method Low–Medium

Methodological evidence Medium

Empirical evidence High

Current state of the art Very High

This prevents foundational papers from being

incorrectly flagged. **4. Outdated Citation**

Risk Score

The proposed system combines the preceding signals into an **Outdated Citation Risk Score (OCRS)**:

$$\begin{aligned} &[\\ \text{OCRS} = & \\ &w_1T + w_2S + w_3N + w_4I - w_5F \\ &] \end{aligned}$$

where:

- (T) = temporal risk;
- (S) = supersession evidence;
- (N) = newer-literature evidence;
- (I) = citation-intent risk;
- (F) = foundational importance.

The output can be categorized as:

Score Classification

0–0.24 Current

0.25–0.49 Low risk

0.50–0.69 Potentially outdated

0.70–0.84 Likely superseded

0.85–1.00 High risk

These thresholds are proposed for experimental evaluation and should ultimately be learned and calibrated using expert annotations.

5. Current Approaches and Developments

Current AI literature systems increasingly combine retrieval with language generation. RAG provides a mechanism for accessing external information rather than relying exclusively on knowledge stored in model parameters [1]. ([DOI](#))

Scientific claim-verification systems such as SciFact demonstrate that AI can retrieve supporting or contradicting scientific evidence [5]. ([ACL Anthology](#)) Large scholarly corpora such as S2ORC provide the structured metadata and citation relationships needed for large-scale analysis [7]. ([arXiv](#))

Nevertheless, these approaches generally emphasize **retrieval, relevance, or claim support** rather than explicit citation aging. The proposed pipeline therefore acts as an additional verification layer after an AI system generates its recommendations.

6. Research Challenges and Limitations

Several challenges must be addressed.

First, outdatedness is context-dependent. A paper may be old but still be the correct original source for a historical or methodological statement.

Second, scientific fields evolve at different rates. A fixed five-year or ten-year threshold cannot be applied universally.

Third, citation networks contain biases. Citation counts are influenced by field size, visibility, language, and publication practices and should not be interpreted as direct measures of scientific quality.

Fourth, semantic similarity does not necessarily indicate supersession. A newer paper may complement rather than replace an older study.

Finally, bibliographic databases may contain incomplete or inconsistent metadata. Duplicate records, preprints, journal versions, and missing identifiers can complicate automated matching.

7. Research Gaps and Future Directions

A major research gap is the lack of a standardized benchmark specifically for **outdated citation detection in AI-generated literature recommendations**.

Future research should develop datasets containing expert labels such as:

- current;
- foundational but old;
- potentially outdated;
- superseded;
- retracted;
- irrelevant.

Temporal evaluation should also be used. Models should be trained on earlier literature and tested on later publications to determine whether they can recognize genuine changes in the evidence base.

Future systems should also move from paper-level to **claim-level monitoring**:

[
Claim $\rightarrow$ Citation $\rightarrow$ New Evidence
]

Such systems could continuously monitor a researcher's bibliography and notify them when important new evidence appears.

Another promising direction is human-in-the-loop learning. Researchers could confirm or reject warnings, allowing the system to learn discipline-specific citation practices.

8. Conclusion

AI systems are becoming increasingly useful for scientific literature discovery and synthesis, but reliable citation generation requires more than finding papers that are relevant and authentic. Scientific knowledge evolves, and older references may become inadequate for contemporary claims.

This paper proposed an automated pipeline that combines bibliographic verification, temporal analysis, newer-literature retrieval, semantic similarity, citation-intent classification, and supersession detection. The proposed Outdated Citation Risk Score provides a mechanism for prioritizing citations that require human review.

The central principle is that **old does not necessarily mean outdated**.

Foundational papers should be preserved when they remain appropriate, while citations supporting rapidly changing empirical or state-of-the-art claims should be periodically reassessed.

The proposed approach provides a foundation for future AI research assistants capable not only of recommending literature but also of continuously checking whether the literature they recommend remains current, appropriate, and scientifically defensible.

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